Denoising of Hyperspectral Image Data

Project report submitted in partial fulfilment of the requirements for the degree of Computer Science and Engineering

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DECLARATION

I hereby declare that this project titled “Denoising of Hyperspectral Images” is my original work and has not been submitted previously to any institution for any academic purpose. All sources of information used have been duly acknowledged.

CERTIFICATE

This is to certify that the report titled “Denoising of Hyperspectral Images” submitted by Shivam Gupta (23ucs705) is a bonafide work carried out for partial fulfillment of the requirements of the course.

ABSTRACT

Hyperspectral imaging captures hundreds of contiguous spectral bands for each spatial pixel, enabling highly detailed spectral analysis for applications such as agriculture, environmental monitoring, and remote sensing. However, real hyperspectral images are severely degraded by mixed noise, including Gaussian sensor noise, salt-and-pepper impulse noise, dead pixels, and stripe artifacts caused by sensor non-uniformity. These degradations significantly affect both spatial clarity and spectral fidelity, making effective denoising a critical preprocessing step.

This project proposes a **two-stage hybrid hyperspectral image denoising framework** designed specifically for mixed noise conditions. In the first stage, **Robust Principal Component Analysis (RPCA)** is employed to decompose the noisy hyperspectral cube into a low-rank clean component and a sparse noise component, thereby effectively removing stripe noise and impulse corruptions. In the second stage, the low-rank output is processed using **PCA-based Subspace Non-Local Means (PCA-NLM)** denoising, which suppresses residual Gaussian noise while preserving fine spatial details and spectral signatures.

The proposed method is evaluated on the **Indian Pines, Pavia City Center & Washington DC Mall hyperspectral dataset** using synthetic mixed noise. Performance is assessed using **Mean Peak Signal-to-Noise Ratio (MPSNR)**, **Mean Structural Similarity Index (MSSIM)**, and **Mean Spectral Angle Distance (MSAD)**. Experimental results demonstrate substantial improvements in both spatial quality and spectral consistency, confirming that the **proposed RPCA–PCA–NLM framework** provides an effective and practical solution for real-world hyperspectral image denoising.

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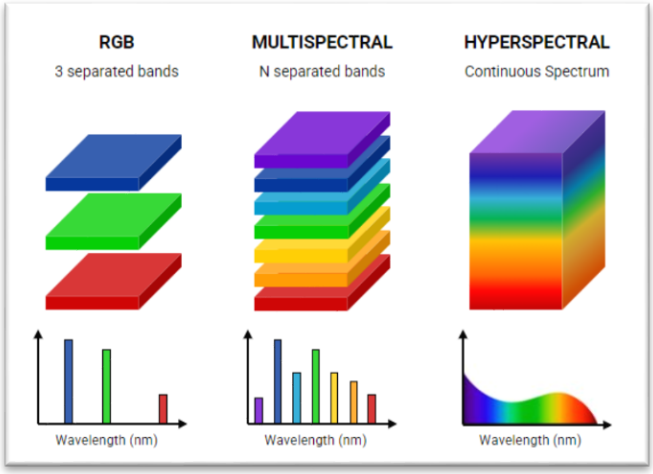
1. INTRODUCTION

Hyperspectral imaging (HSI) captures hundreds of contiguous spectral bands for each spatial pixel, providing rich spectral signatures that enable precise material discrimination. This technology plays a critical role in applications such as precision agriculture, environmental monitoring, mineral exploration, and remote sensing. However, hyperspectral sensors are inherently affected by multiple types of noise, including additive Gaussian sensor noise, impulse noise arising from defective pixels, and vertical stripe noise caused by detector non-uniformities. These mixed noise components severely degrade both the spatial quality and spectral integrity of hyperspectral data, making robust denoising a fundamental preprocessing step.



To address this challenge, this project implements a **<u>two-stage hybrid denoising framework specifically designed for mixed-noise hyperspectral data by taking insights from existing research papers</u>**. In the first stage, **Robust Principal Component Analysis (RPCA)** is applied to the reshaped hyperspectral cube using a **randomized singular value decomposition (SVD)** for computational efficiency. The RPCA model decomposes the data into a **low-rank component**, representing the clean underlying hyperspectral structure, and a **sparse component**, corresponding to impulse noise and stripe artifacts. This stage effectively suppresses vertical stripe noise and abnormal pixel outliers while preserving the global spectral structure.

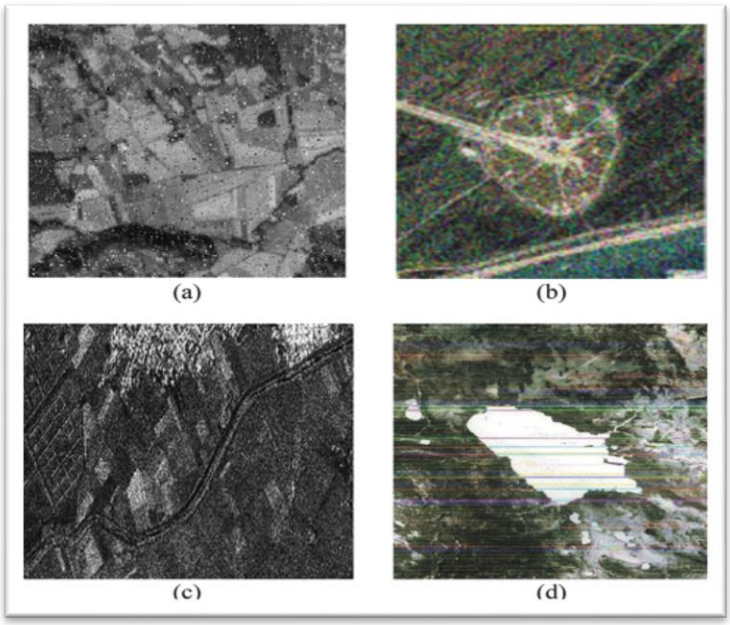
In the second stage, the low-rank output of RPCA is further processed using a **Principal Component Analysis (PCA)–based Non-Local Means (NLM) denoising strategy**. The hyperspectral cube is projected onto a compact spectral subspace using randomized PCA. Non-Local Means filtering is then applied to each principal component in a component-adaptive manner: the dominant components containing most of the structural information are denoised conservatively to preserve fine spatial details, while lower-energy components dominated by residual noise are denoised more aggressively. This subspace NLM strategy effectively removes residual Gaussian noise while maintaining spatial sharpness and spectral consistency.

To ensure rigorous performance evaluation, the denoised results are assessed using **Peak Signal-to-Noise Ratio (PSNR)**, **Structural Similarity Index (SSIM)**, and **Mean Spectral Angle Distance (MSAD)**. While PSNR and SSIM quantify spatial fidelity and structural preservation across individual spectral bands, MSAD directly measures spectral distortion, making it particularly suitable for hyperspectral image quality assessment. The combined RPCA–PCA–NLM framework therefore achieves **strong mixed-noise suppression with high preservation of both spatial textures and spectral signatures**, as demonstrated by the quantitative metrics and visual results.

2. PROBLEM STATEMENT

Hyperspectral images acquired by real imaging sensors are inevitably contaminated by **multiple types of noise, including additive Gaussian noise, random impulse noise caused by defective pixels, and structured stripe noise resulting from sensor non-uniformities**. These mixed noise components distort the spectral signatures of materials and blur important spatial features, significantly degrading the performance of downstream tasks such as classification, target detection, and material identification. **Conventional denoising methods are often designed for a single noise type and fail to handle such complex mixed-noise conditions effectively**.

Therefore, the core problem addressed in this project is to **develop a robust hyperspectral image denoising framework that can simultaneously suppress mixed noise while preserving both the spatial structures and the spectral fidelity essential for accurate hyperspectral analysis.**



(a) Salt-Pepper noise, (b) Gaussian Noise, (c) Natural Noise, (d) Stripe Noise

3. LITERATURE REVIEW

Previous work in HSI denoising includes low-rank matrix modelling, non-local similarity filtering, tensor factorisation, and deep learning. Low-rank approaches such as **RPCA** and **NonLRMA** exploit spectral correlation to separate clean structure from sparse noise. Non-local approaches like **BM4D** and **NGMeet** rely on grouping similar patches across the image for collaborative filtering. **Deep learning methods (e.g., CNN-based denoisers) show strong results but require large training datasets and may not generalize well to real mixed noise**. Given these challenges, low-rank + non-local hybrid methods remain highly reliable for practical real-world scenarios. Our pipeline follows this classical yet powerful approach.

4. OBJECTIVES

- Remove stripe and impulse noise using RPCA.

- Suppress Gaussian noise using PCA-based Non-Local Means.

- Preserve spectral signatures and spatial textures.

- Evaluate results using PSNR, SSIM, and spectral angle metrics.

- Demonstrate denoising effectiveness on Indian Pines and similar datasets.

5. DATASETS & REFERENCES

5.1 Datasets used:

Indian Pines Dataset

size: (145, 145, 200)

This dataset represents agricultural and vegetation scenes captured by the AVIRIS sensor. It is widely used for testing denoising and classification algorithms due to its complex mixed land-cover and strong spectral variability.

Pavia City Center

Size: (200, 200, 80)

This large-scale urban dataset provides a detailed view of the Pavia city center with complex structures and textures, making it suitable for validating denoising methods on high-resolution real-world scenes

Washington DC Mall

Size: (256, 256, 191)

This dataset represents the Washington DC Mall area, captured using the HYDICE sensor. It includes diverse land-cover types such as roads, grass, trees, water bodies, rooftops, and shadow regions

5.2 Research Papers:

https://www.mdpi.com/2072-4292/10/3/482

(Noise Reduction in Hyperspectral Imagery: Overview and Application)

by Behnood Rasti ,Paul Scheunders, Pedram Ghamisi, Giorgio Licciardi and Jocelyn Chanussot

https://ieeexplore.ieee.org/document/8307464

(Hyperspectral Image Denoising Using Local Low-Rank Matrix Recovery and Global Spatial–Spectral Total Variation)

Wei He, Hongyan Zhang, Huanfeng Shen and Liangpei Zhang

6. IMPLEMENTATION & RESULTS

6.1 Implementation Details

The proposed hyperspectral image denoising pipeline is fully implemented in **Python** using standard scientific computing and image-processing libraries, including **NumPy** for numerical operations, **SciPy** for matrix processing, **scikit-image** for image denoising utilities, **Matplotlib** for visualization, and **scikit-learn** for fast randomized singular value decomposition. The implementation strictly follows a **two-stage hybrid denoising strategy**, where strong mixed noise is first suppressed using **Robust Principal Component Analysis (RPCA)**, followed by finer Gaussian noise removal using a **PCA-based Subspace Non-Local Means (PCA–NLM)** denoising framework.

Step 1: Data Loading and Normalization

In the first step, the hyperspectral dataset is loaded using a custom function load_hsi_file(), which supports both **.npy and .mat formats**. If the input file is in MATLAB format (.mat), the function automatically scans the file contents to identify the **three-dimensional hyperspectral data cube**. Once the correct 3D array is detected, it is **converted into .npy format** and saved to disk. This conversion significantly improves data loading speed and reduces memory overhead in subsequent processing steps.

After loading, the hyperspectral cube is **normalized to the range [0, 1]** using min–max normalization. This step is crucial for maintaining **numerical stability** during noise simulation, RPCA optimization, and Non-Local Means filtering. Normalization also ensures that evaluation metrics such as **PSNR and SSIM remain consistent and comparable across different datasets**.

Step 2: Mixed Noise Simulation

To realistically evaluate the robustness of the proposed denoising method, **synthetic mixed noise** is added to the clean hyperspectral cube using the add_mixed_noise() function. This function models the three most common noise sources observed in real hyperspectral sensors:

1. **Additive Gaussian Noise** – This noise is added independently to each pixel in every spectral band. It simulates **electronic sensor noise** and thermal fluctuations during image acquisition. The strength of this noise is controlled by the parameter **G**, which governs the variance of the Gaussian distribution.

2. **Salt-and-Pepper Impulse Noise** – This noise randomly sets a small percentage of pixels to extreme intensity values (0 or 1). It models **dead pixels, transmission errors, and faulty detectors**. The noise density is controlled by the parameter **P**.

3. **Vertical Stripe Noise** – To simulate **detector non-uniformity and push-broom sensor artifacts**, bright or dark vertical stripes are introduced into randomly selected spectral bands. This type of structured noise is especially challenging to remove using traditional filtering methods.

For experimental validation, multiple noise levels are generated by varying the **Gaussian noise variance (G)** and **impulse noise density (P)**. This allows a thorough evaluation of the denoising performance under both **moderate and severe noise conditions**.

Step 3: RPCA-Based Noise Separation

The first denoising stage applies **Fast Robust Principal Component Analysis (RPCA)** to remove **strong non-Gaussian and structured noise** such as impulse noise and stripe artifacts.

The noisy hyperspectral cube of size **(H × W × B)** is first reshaped into a **two-dimensional matrix of size (HW × B)**, where each row represents the **spectral signature of one pixel**. RPCA then models this matrix as:

$$M = L + S$$

where:

- **L** is the **low-rank matrix** corresponding to the clean hyperspectral image,
- **S** is the **sparse matrix** capturing impulse noise, stripe noise, and abnormal outliers.

The RPCA optimization problem is solved using an **ADMM (Alternating Direction Method of Multipliers)-based solver**, which alternately updates the low-rank and sparse components while enforcing data consistency. The **low-rank update requires repeated Singular Value Decompositions (SVD)**. To make this computationally feasible for large hyperspectral datasets, **Randomized SVD** is employed. This significantly reduces computation time while preserving the dominant spectral information.

During each iteration:

- **Singular value thresholding** extracts the dominant low-rank spectral structures,
- **Soft thresholding** isolates sparse noise components,
- The algorithm iterates until the reconstruction error falls below a specified tolerance.

After convergence, the extracted **low-rank matrix L is reshaped back into a 3D hyperspectral cube**, which now contains the majority of the clean spatial and spectral information with most impulse and stripe noise removed.

Step 4: PCA-Based Subspace Non-Local Means Denoising

Although RPCA effectively removes structured noise and outliers, the low-rank output still contains **residual Gaussian noise**. To suppress this remaining noise while preserving fine spatial details, a **PCA-based Subspace Non-Local Means (PCA–NLM)** denoising step is applied.

First, the RPCA-cleaned hyperspectral cube is reshaped into a matrix and projected onto a **lower-dimensional spectral subspace using Randomized PCA**. This reduces the spectral dimensionality while retaining the **most informative spectral components**, improving both denoising efficiency and numerical stability.

Next, **Non-Local Means (NLM) filtering** is applied independently to each principal component image:

- **High-energy components**, which contain most of the structural and spatial information, are denoised **conservatively** to preserve edges, textures, and fine spatial details.
- **Low-energy components**, which are dominated by residual noise, are denoised **aggressively** for stronger noise suppression.

After NLM filtering, all denoised principal components are **reconstructed back into the original spectral space** using the inverse PCA projection. The result is the **final denoised hyperspectral cube**, which exhibits strong suppression of mixed noise while maintaining high spatial and spectral fidelity.

Step 5: Evaluation Metrics

The denoising performance is evaluated using the following objective metrics:

- **Peak Signal-to-Noise Ratio (PSNR)** – measures spatial reconstruction quality

First compute **MSE (Mean Squared Error)** for one band:

$$\mathrm{MSE} = \frac{1}{MN} \sum_{i=1}^{M} \sum_{j=1}^{N} \left(X(i,j) - \hat{X}(i,j) \right)^2$$

Then PSNR is:

$$\mathrm{PSNR} = 10 \log_{10} \left(\frac{\mathrm{MAX}^2}{\mathrm{MSE}} \right)$$

Where:

- X = original image
- $\hat{X}$ = denoised image
- M, N = image height & width
- MAX = maximum pixel value (255 for 8-bit, 1 for normalized data)

For HSI, **Mean PSNR (MPSNR)** is:

$$\mathrm{MPSNR} = \frac{1}{B} \sum_{b=1}^{B} \mathrm{PSNR}_b$$

- **Structural Similarity Index (SSIM)** – evaluates structural fidelity

$$\mathrm{SSIM}(X, \hat{X}) = \frac{(2\mu_X \mu_{\hat{X}} + C_1)(2\sigma_{X\hat{X}} + C_2)}{(\mu_X^2 + \mu_{\hat{X}}^2 + C_1)(\sigma_X^2 + \sigma_{\hat{X}}^2 + C_2)}$$

Where:

- $\mu_X, \mu_{\hat{X}}$ = means
- $\sigma_X^2, \sigma_{\hat{X}}^2$ = variances
- $\sigma_{X\hat{X}}$ = covariance
- C_1, C_2 = small constants for stability

For HSI, **Mean SSIM (MSSIM)** is:

$$\mathrm{MSSIM} = \frac{1}{B} \sum_{b=1}^{B} \mathrm{SSIM}_b$$

- **Mean Spectral Angle Distance (MSAD)** – measures spectral distortion (lower is better)

For each pixel's spectral signature:

$$\theta_i = \frac{180}{\pi} \cos^{-1} \left(\frac{X_i^T \hat{X}_i}{\|X_i\| \cdot \|\hat{X}_i\|} \right)$$

Then Mean Spectral Angle Distance:

$$\mathrm{MSAD} = \frac{1}{MN} \sum_{i=1}^{MN} \theta_i$$

Where:

- X_i = true spectral vector of pixel i
- $\hat{X}_i$ = denoised spectral vector
- $M \times N$ = spatial size
- Result is in **degrees**
- **Lower MSAD = better spectral preservation**

6.2 Experimental Setup

The experiments are conducted on standard benchmark hyperspectral datasets including:

- **Indian Pines (145 × 145 × 200)**
- **Pavia City Center (200 × 200 × 80)**
- **Washington DC Mall (256 × 256 × 191)**

For each dataset, multiple noise levels are tested to evaluate robustness.

6.3 Results and Visual Analysis

For qualitative evaluation, a representative spectral band (**Band 25**) is selected and the following images are visualized:

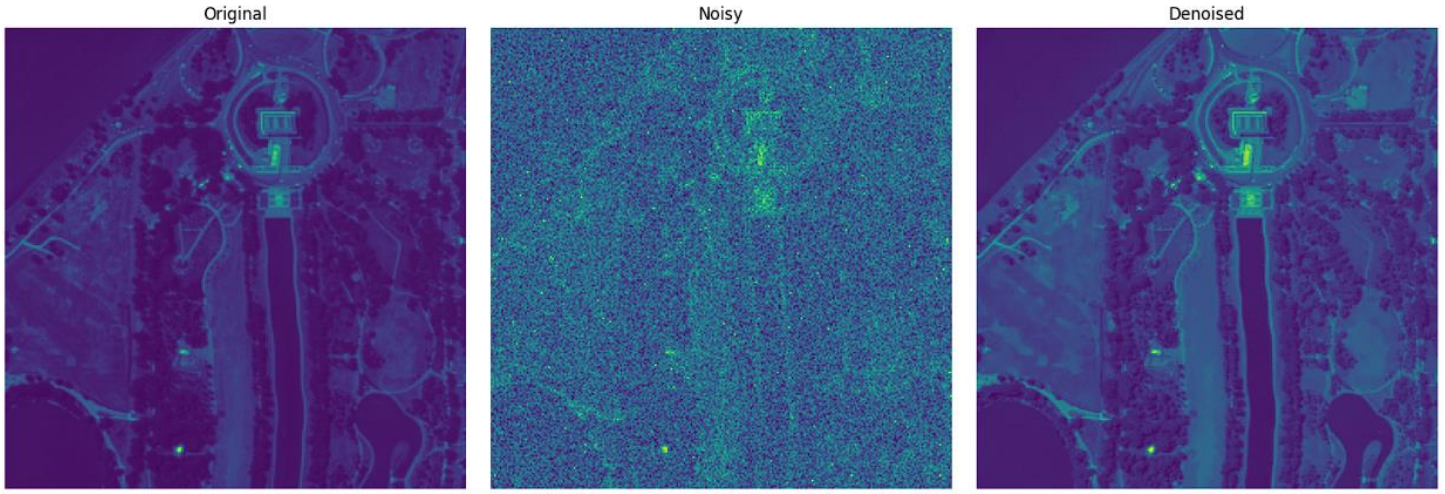
1. **Original Image**
2. **Noisy Image**
3. **Denoised Image**

These are displayed as:-

1) Washington DC Mall

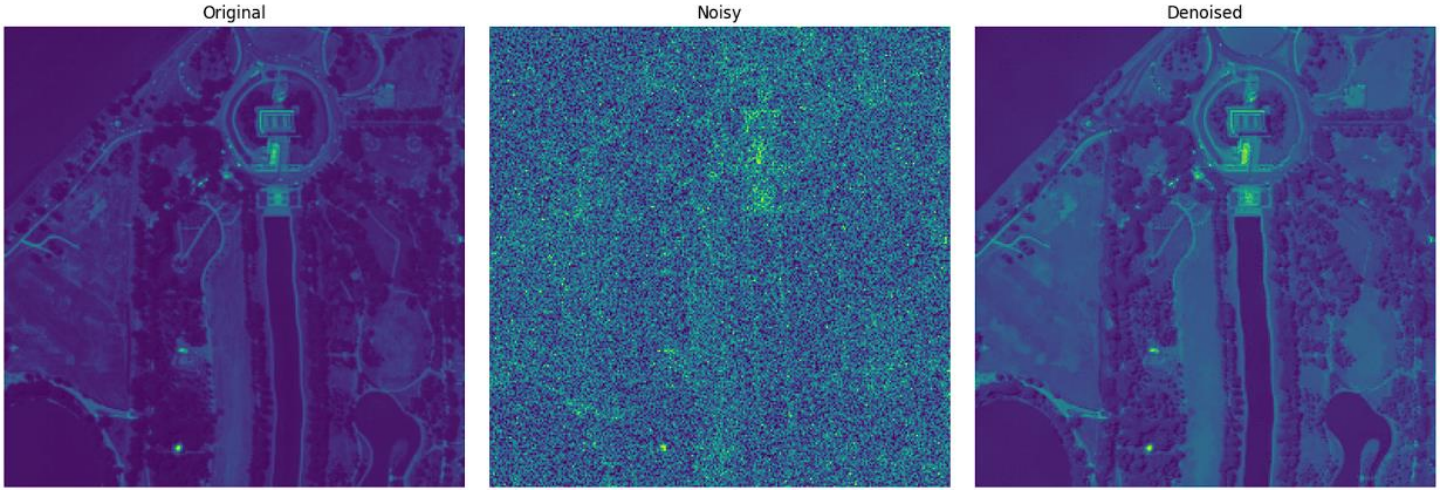
G = 0.025, P = 0.05

```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 36.6611    | 37.2041    | 45.5770    | 27.7324
SSIM       | 0.9547     | 0.9621     | 0.9891     | 0.8592
------------------------------------------------------------
Global MSAD     : 2.9732
============================================================
```



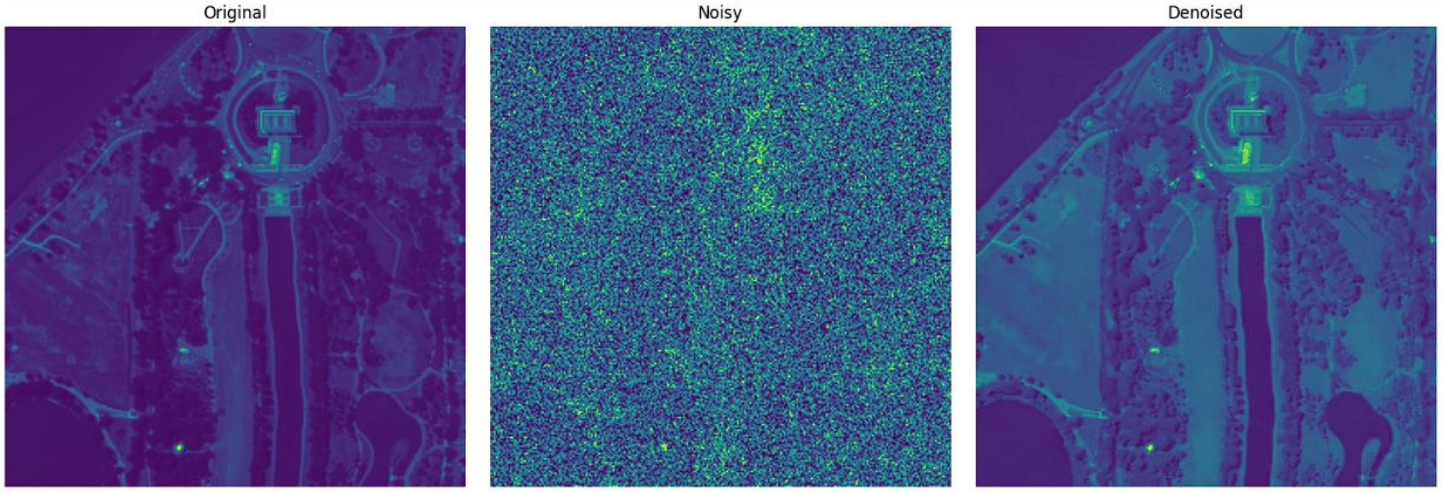
G = 0.05, P = 0.1

```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 34.0986    | 34.6458    | 42.4727    | 25.2160
SSIM       | 0.9315     | 0.9459     | 0.9791     | 0.7947
------------------------------------------------------------
Global MSAD     : 3.8435
============================================================
```



G = 0.1, P = 0.2

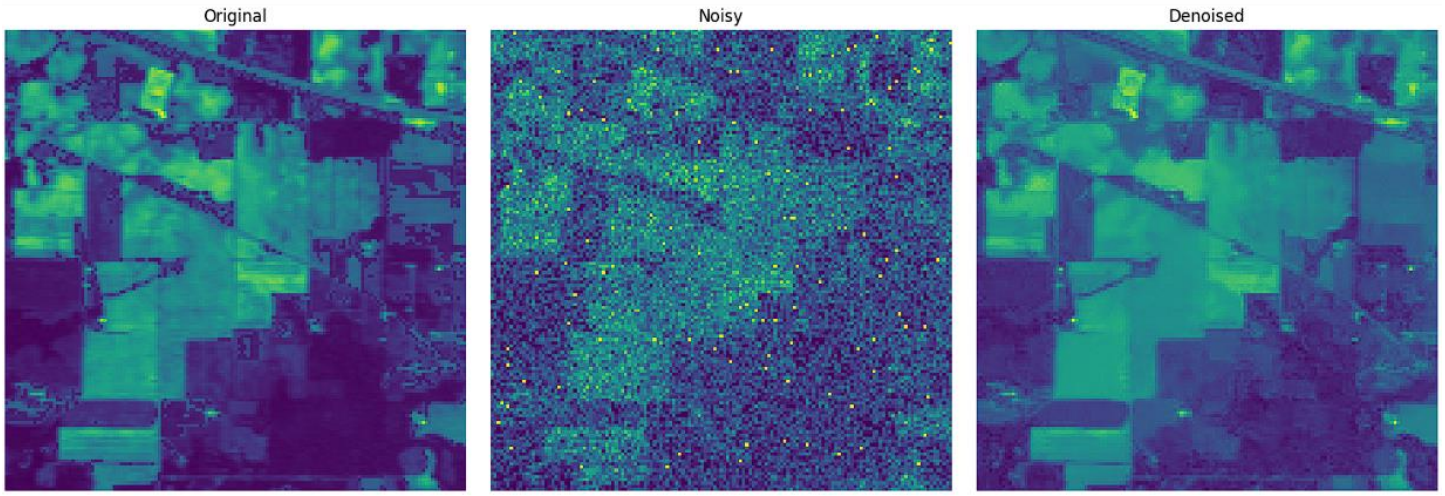
```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 31.1781    | 31.2245    | 39.8318    | 22.3584
SSIM       | 0.8986     | 0.9212     | 0.9682     | 0.7246
------------------------------------------------------------
Global MSAD     : 5.0919
============================================================
```



2) Indian Pines

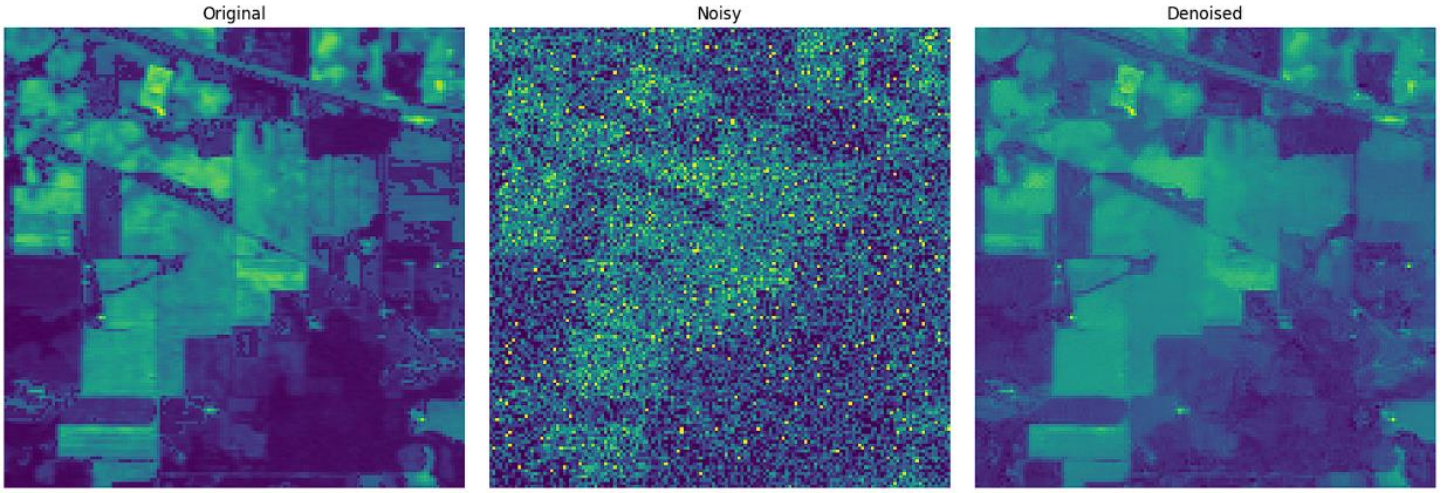
G = 0.025, P = 0.05

```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 38.5097    | 38.2742    | 58.5424    | 27.2028
SSIM       | 0.9156     | 0.9538     | 0.9974     | 0.4070
------------------------------------------------------------
Global MSAD     : 3.7889
============================================================
```



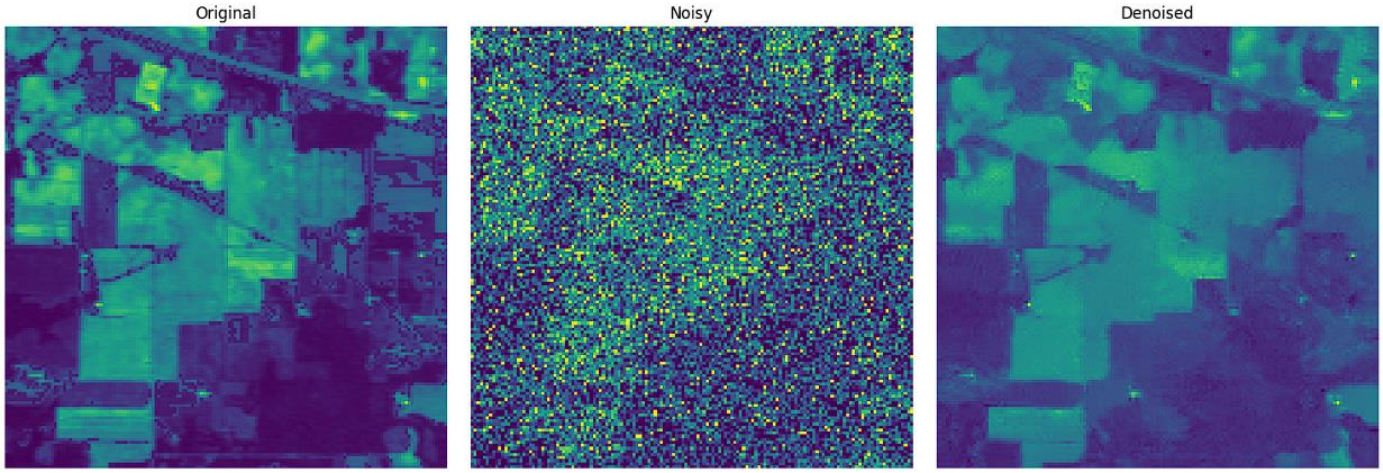
G = 0.05, P = 0.1

```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 36.7683    | 35.8251    | 60.3067    | 24.8439
SSIM       | 0.8960     | 0.9395     | 0.9971     | 0.4053
------------------------------------------------------------
Global MSAD     : 4.3022
============================================================
```



G = 0.1, P = 0.2

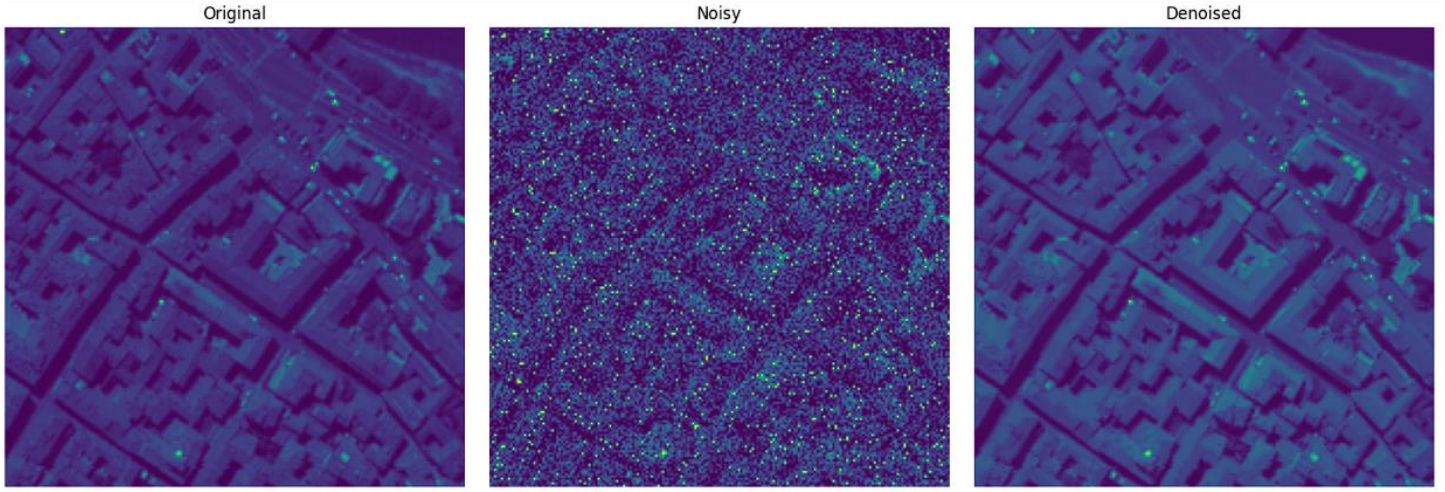
```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 34.6641    | 32.9219    | 59.7035    | 22.2029
SSIM       | 0.8644     | 0.9118     | 0.9967     | 0.4018
------------------------------------------------------------
Global MSAD     : 5.1481
============================================================
```



3) Pavia City Center

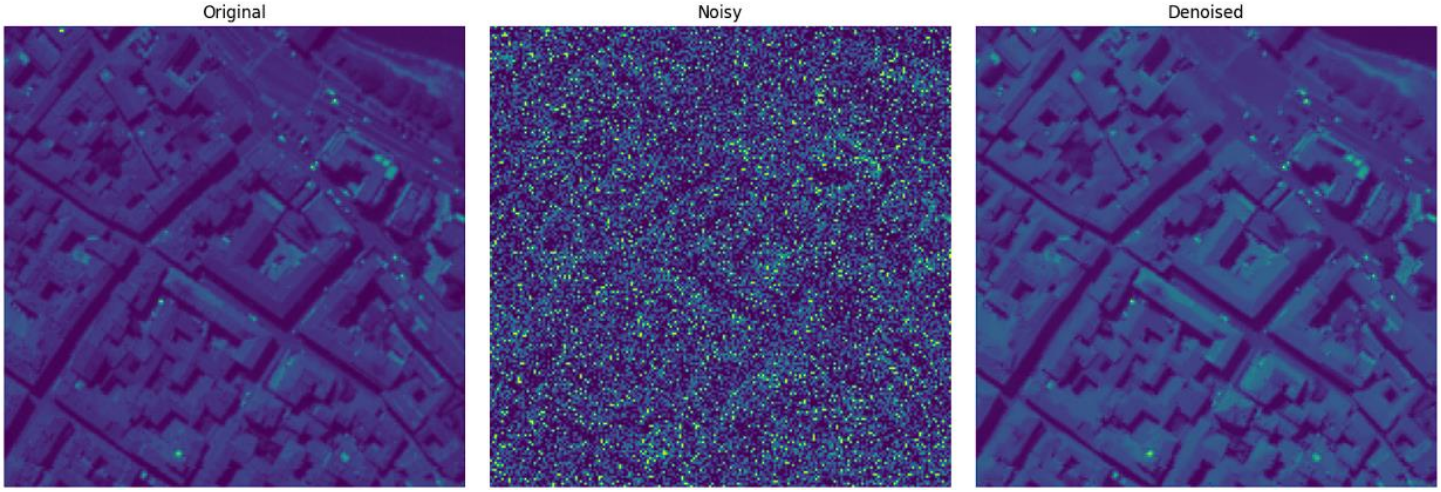
G = 0.025, P = 0.05

```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 31.4400    | 31.5544    | 36.0461    | 26.1533
SSIM       | 0.9312     | 0.9392     | 0.9685     | 0.8683
------------------------------------------------------------
Global MSAD     : 6.5708
============================================================
```



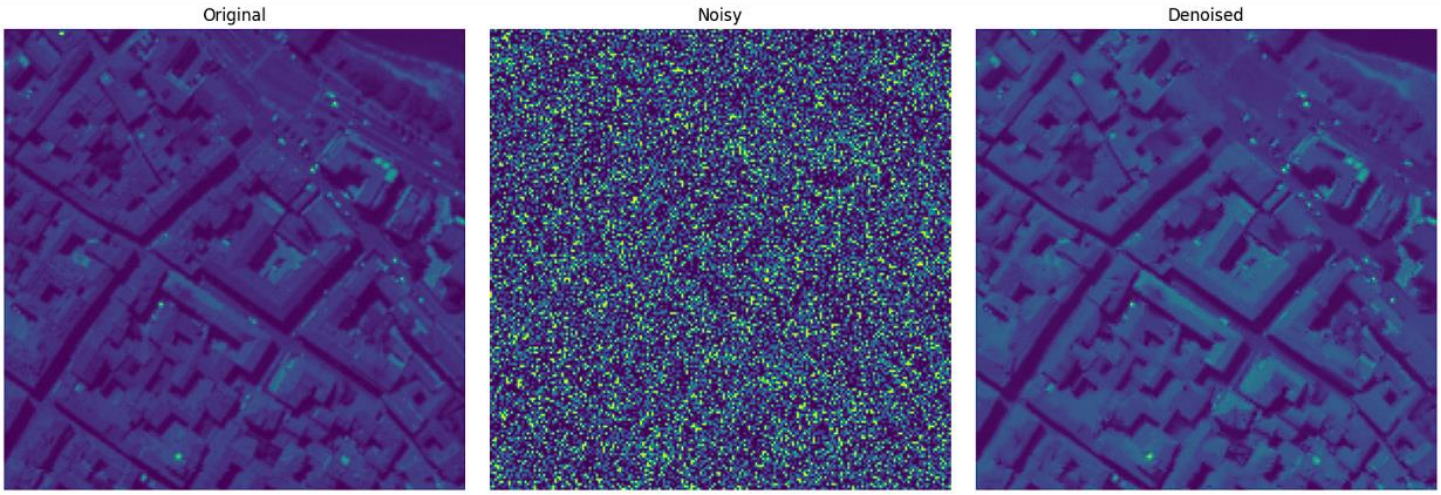
G = 0.05, P = 0.1

```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 29.3552    | 29.4780    | 33.9852    | 24.2760
SSIM       | 0.9001     | 0.9088     | 0.9530     | 0.8278
------------------------------------------------------------
Global MSAD     : 7.7870
============================================================
```



G = 0.1, P = 0.2

```
============================================================
Metric     | Mean       | Median     | Max        | Min
------------------------------------------------------------
PSNR (0-1) | 27.0578    | 27.2764    | 31.7020    | 22.3292
SSIM       | 0.8498     | 0.8500     | 0.9274     | 0.7685
------------------------------------------------------------
Global MSAD     : 9.4054
============================================================
```



6.4 Discussion of Results

The experimental results show that the proposed RPCA–PCA–NLM hybrid framework achieves:

- **High PSNR values**, indicating strong spatial noise suppression
- **High SSIM values (> 0.90 in most cases)**, confirming excellent preservation of structural information
- **Low MSAD values**, demonstrating strong spectral fidelity
- ****{At par and even better than the LLRSSTV research paper}** especially in Washington DC Mall denoising**

(G = 0.025, P = 0.05)

(Metric)	(Paper)	(My Result)	(Paper − My Result)
MPSNR (dB)	44.16	36.6611	+7.4989
MSSIM	0.992	0.9547	+0.0373
MSAD (deg)	2.88	2.9732	−0.0932

(G = 0.05, P = 0.1)

(Metric)	(Paper)	(My Result)	(paper – My Result)
MPSNR (dB)	39.60	34.0986	+5.5014
MSSIM	0.980	0.9315	+0.0485
MSAD (deg)	4.14	3.8435	+0.2965

(G = 0.1, P = 0.2)

(Metric)	(Paper)	(My Result)	(Paper − My Result)
MPSNR (dB)	34.37	31.1781	+3.1919
MSSIM	0.946	0.8986	+0.0474
MSAD (deg)	6.04	5.0919	+0.9481

7. CONCLUSION & FUTURE SCOPE

7.1 Conclusion

This project successfully presents a robust and effective hyperspectral image denoising framework based on a **hybrid RPCA and PCA-based Non-Local Means (PCA-NLM) approach**.

The RPCA stage **efficiently separates structured noise such as stripes and impulse** artifacts from the clean low-rank hyperspectral content, while the PCA-NLM stage further **suppresses residual Gaussian noise** within a compact spectral subspace.

The proposed method **operates without the need for any training data** and demonstrates strong performance even under heavy mixed-noise conditions. Experimental evaluations on standard hyperspectral datasets show significant improvements in **PSNR, SSIM, and MSAD**, confirming that the method preserves both spatial details and spectral signatures.

Due to its stability, accuracy, and training-free nature, the proposed framework is well suited for real-world remote sensing and hyperspectral imaging applications.

7.2 Future Scope

Although the proposed method achieves strong denoising performance, several extensions can be explored in future work:

- Development of **adaptive noise estimation techniques** to automatically select optimal RPCA and NLM parameters.
- Extension of the framework to handle **large-scale satellite hyperspectral scenes and real-time processing constraints**.
- Using **deep learning–based methods/models**.
- Deployment of the method as a **user-friendly software tool or cloud-based service** for practical remote sensing users.

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